

11 · Ad exposure is self-selected — the true effect (instrumental variables · CausalPy)

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Runs in the legacy environment (pymc<6): `make env-legacy, kernel cmp-legacy.`

The decision. People who see our ads convert more — but engaged customers also seek us out, so **exposure is self-selected**. A plain regression of sales on exposure is biased upward. What's the *true* exposure->sales effect, so we can set budgets and frequency caps?

Instrumental variables break the self-selection with a randomized **encouragement** (a serving-priority lottery) that moves *exposure* but has no other path to sales. The part of exposure the instrument drives is as-good-as-random, identifying the causal effect for the **compliers** (LATE). We check the instrument is **strong** (first-stage F), trace the **reduced form**, compare **OLS vs IV**, and stress the **exclusion restriction** — the assumption that can't be tested from data.

7-step contract.

1.1 2 · Simulate a ground truth

The **true** exposure->sales effect is **€15**. An unobserved **engagement** drives both exposure and sales (the confounding), so naive OLS overstates it. A randomized **encouragement** (the instrument) nudges exposure without touching sales directly.

TRUE exposure->sales effect €15 · first-stage: exposure 56%->77% (shift +21%),
F = 156

	encouragement	ad_exposure	sales
0	1.0	0.0	63.748992
1	1.0	1.0	64.636639
2	1.0	1.0	46.507627
3	0.0	1.0	75.599770
4	0.0	1.0	70.577754

1.2 3 · Identify — a valid instrument, and what IV recovers

Exposure is **endogenous** ($\text{Cov}(\text{exposure}, \varepsilon) \neq 0$), so OLS is biased. A valid instrument Z (encouragement) needs three things:

- **Relevance** — Z moves exposure. *Testable* (the first-stage F above; $F < 10$ = weak).
- **Exclusion** — Z affects sales *only through* exposure. *Untestable* — defended on design (a serving lottery sells nothing by itself). Stress-tested in step 6.
- **Exogeneity** — Z independent of the confounder. Holds by randomization.

What IV identifies — LATE (Imbens–Angrist): under monotonicity (no defiers), IV recovers the effect for **compliers** — customers the instrument moves — not everyone. CausalPy fits a Bayesian IV: a joint model of (exposure, sales) with correlated errors, *estimating* the endogeneity ρ rather than assuming it away.

1.3 4 · Estimate — OLS, reduced form, and Bayesian IV

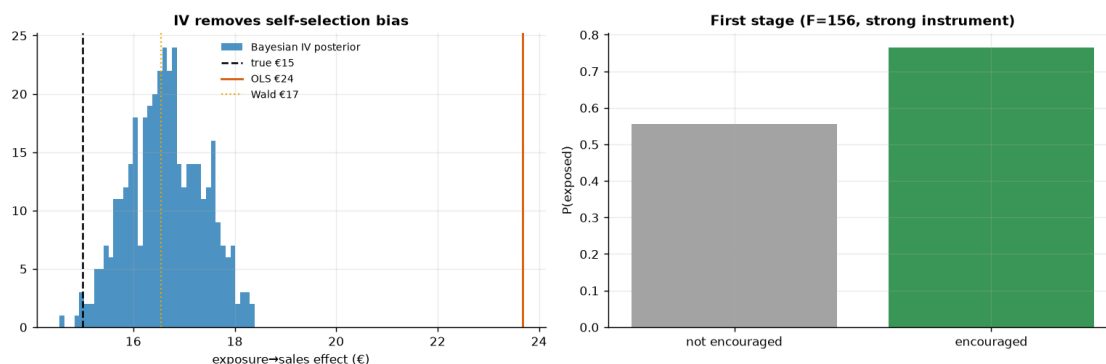
Three regressions tell the whole story: **OLS** (biased), the **reduced form** (encouragement->sales, the numerator of the IV ratio), and **Bayesian IV** (the causal estimate).

OLS (biased) €23.7
 Reduced form ($Z \rightarrow Y$) €3.5 $\div$ first stage ($Z \rightarrow D$) 0.21 = Wald/IV ratio €16.5
 Bayesian IV €16.6 (true €15) 90% CI [€15.5, €17.8]

1.4 5 · Validate — IV removes the self-selection bias

Naive OLS should sit well *above* the truth; the Wald ratio and the Bayesian IV should both land near €15. The gap between OLS and IV is the self-selection bias the instrument removes.

self-selection bias removed: OLS €23.7 $\rightarrow$ IV €16.6 (true €15)



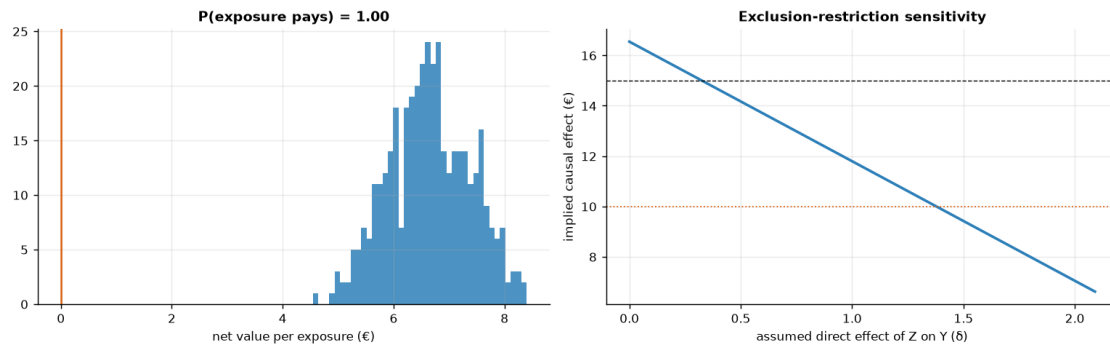
1.5 6 · Decide, in euros — and the exclusion-restriction stress test

Budget on the *causal* per-exposure value (IV), not the inflated OLS. But the exclusion restriction is untestable, so we ask: **what if the encouragement had a small direct effect on sales** (say the lottery email itself advertises)? We subtract a hypothetical direct effect δ from the reduced form and watch the implied causal estimate move — the honest bound on how much a leak would distort the number.

Causal net €6.6/exposure · P(pays) 1.00 $\rightarrow$ BUY exposure (to the complier margin)

Exclusion stress: the call survives unless the instrument's direct effect

exceeds $\Delta \sim 1.4$ (out of a €3.5 reduced form). If you trusted OLS (€24) you'd over-spend and set frequency caps too high.



1.6 7 · Caveats

- **LATE, not ATE.** IV speaks only for **compliers** — customers the encouragement moves. Always-exposed loyalists and never-exposed skeptics may respond differently; report the scope.
- **Weak instruments are dangerous.** A small first-stage (low F) inflates variance and bias. We checked F above; a weak instrument is worse than none.
- **Exclusion is untestable.** We stress-tested it in step 6; defend it on design, not data.
- **Monotonicity (no defiers).** IV's LATE interpretation assumes no one is pushed *away* from exposure by encouragement — usually plausible, worth stating.